

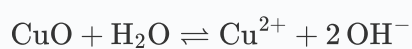
Question

Bright green **A** ($\omega_{\text{Cu}} = 57.5\%$) and blue **B** ($\omega_{\text{Cu}} = 55.3\%$) are widely found in nature, and these two minerals often coexist. Heating a mixture of copper sulfate solution and sodium carbonate in a sealed tube to 150–170 °C yields **A** (**Reaction 1**), while a lower reaction temperature produces **B** (**Reaction 2**). ($\omega\%$ represents mass fraction.)

Heating **A** or **B** within a certain temperature range decomposes them to produce copper oxide. Take 7.96 mg of CuO in a beaker, add 100 mL of deionized water, and stir continuously while adding solid $(\text{NH}_4)_2\text{CO}_3$. Assuming the solution volume remains constant at 100 mL, calculate the minimum mass of $(\text{NH}_4)_2\text{CO}_3$ required to fully dissolve the solid (assuming no other basic salt precipitates form and no CO_2 escapes the system). Also, determine the chemical equations for **Reaction 1** and **Reaction 2**.

Given conditions:

Dissolution equilibrium:



Equilibrium constant:

$$K_s = 2.5 \times 10^{-21}$$

Solubility product constants:

- $\text{Cu}(\text{OH})_2$:

$$K_{\text{sp}}(\text{Cu}(\text{OH})_2) = 2.2 \times 10^{-20}$$

- CuCO_3 :

$$K_{\text{sp}}(\text{CuCO}_3) = 1.4 \times 10^{-10}$$

Ionization constants of carbonic acid:

- First ionization:

$$K_{a1}(\text{H}_2\text{CO}_3) = 4.2 \times 10^{-7}$$

- Second ionization:

$$K_{a2}(\text{H}_2\text{CO}_3) = 5.6 \times 10^{-11}$$

Ionic product of water:

$$K_w = 1.0 \times 10^{-14}$$

Acid dissociation constant of ammonium ion:

$$\text{p}K_a(\text{NH}_4^+) = 9.26$$

Cumulative stability constants for $\text{Cu}^{2+} - \text{NH}_3$:

$$\begin{aligned}\lg \beta_1 &= 4.31, \\ \lg \beta_2 &= 7.98, \\ \lg \beta_3 &= 11.02, \\ \lg \beta_4 &= 13.32, \\ \lg \beta_5 &= 12.86.\end{aligned}$$

- A.** In the chemical equation for **Reaction 1**, the sum of the coefficients of the products is 1 greater than the sum of the coefficients of the reactants.
- B.** In the chemical equation for **Reaction 2**, the sum of the coefficients of the products is 1 greater than the sum of the coefficients of the reactants.
- C.** After the solid is completely dissolved, the pH of the solution is approximately 8.20.
- D.** After the solid has just completely dissolved, the amount of ammonia bound to Cu accounts for approximately 0.38 of the total ammonia content (including NH_3 bound to Cu, free NH_3 , and free NH_4^+).
- E.** To dissolve the solid, the minimum mass of $(\text{NH}_4)_2\text{CO}_3$ to be added is approximately 0.27 g.
- F.** To dissolve the solid, the minimum mass of $(\text{NH}_4)_2\text{CO}_3$ to be added is approximately 0.49 g.
- G.** To dissolve the solid, the minimum mass of $(\text{NH}_4)_2\text{CO}_3$ to be added is approximately 0.61 g.
- H.** To dissolve the solid, the minimum mass of $(\text{NH}_4)_2\text{CO}_3$ to be added is approximately 0.85 g.

Answer

Correct Answer: **F**

Detailed Explanation

From the problem description, it is known that **A** and **B** are both basic copper carbonates. Based on the mass fraction of Cu, the chemical formulas of **A** and **B** can be deduced as $\text{Cu}_2(\text{OH})_2\text{CO}_3$ and $\text{Cu}_3(\text{OH})_2(\text{CO}_3)_2$, respectively.

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1 PTS

The molecular formula of **A** is $\text{Cu}_2(\text{OH})_2\text{CO}_3$

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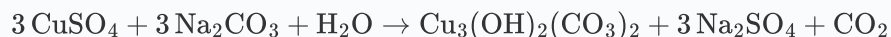
The molecular formula of **B** is $\text{Cu}_3(\text{OH})_2(\text{CO}_3)_2$

Thus, the reactants and products can be determined, and the chemical equations for **Reaction 1** and **Reaction 2** are as follows:

Reaction 1:



Reaction 2:



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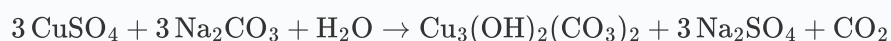
The chemical equation for **Reaction 1** is



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The chemical equation for **Reaction 2** is



Since CuO is highly insoluble, $(\text{NH}_4)_2\text{CO}_3$ is present in large excess compared to CuO . Therefore, it can be assumed that the pH of the solution after the solid dissolves is approximately equal to the pH of a pure $(\text{NH}_4)_2\text{CO}_3$ solution. For a pure $(\text{NH}_4)_2\text{CO}_3$ solution, the charge balance equation is:

$$[\text{NH}_4^+] + [\text{H}^+] = [\text{OH}^-] + [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

Upon analysis, the analytical concentration of $(\text{NH}_4)_2\text{CO}_3$ is greater than $1.0 \times 10^{-3} \text{ mol/L}$, which is significantly higher than H^+ and OH^- . Thus, $[\text{H}^+]$ and $[\text{OH}^-]$ in the above equation can be neglected, simplifying the charge balance equation to:

$$[\text{NH}_4^+] = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

Let the analytical concentration of $(\text{NH}_4)_2\text{CO}_3$ be c . According to the distribution coefficient formulas:

$$[\text{NH}_4^+] = 2c \cdot \frac{[\text{H}^+]}{[\text{H}^+] + K_a(\text{NH}_4^+)}$$

$$[\text{HCO}_3^-] = c \cdot \frac{K_{a1}[\text{H}^+]}{[\text{H}^+]^2 + K_{a1}[\text{H}^+] + K_{a1}K_{a2}}$$

$$[\text{CO}_3^{2-}] = c \cdot \frac{K_{a1}K_{a2}}{[\text{H}^+]^2 + K_{a1}[\text{H}^+] + K_{a1}K_{a2}}$$

By solving these equations, the concentration of H^+ after the solid dissolves is calculated as $[\text{H}^+] = 6.4 \times 10^{-10} \text{ mol/L}$, and $[\text{OH}^-] = 1.6 \times 10^{-5} \text{ mol/L}$.

CHECKPOINT

2 PTS

After the solid dissolves, $[\text{H}^+] = 6.4 \times 10^{-10} \text{ mol/L}$

Since CuO is less soluble than $\text{Cu}(\text{OH})_2$, when the solid just dissolves, the following condition holds:

$$[\text{Cu}^{2+}][\text{OH}^-]^2 = K_s.$$

Thus, the concentration of Cu^{2+} when the solid just dissolves is calculated as $[\text{Cu}^{2+}] = 1.0 \times 10^{-11} \text{ mol/L}$.

CHECKPOINT

1 PTS

When the solid just dissolves, $[\text{Cu}^{2+}] = 1.0 \times 10^{-11} \text{ mol/L}$

Given that $c(\text{Cu}^{2+}) = 1.0 \times 10^{-3} \text{ mol/L}$, the side reaction coefficient for Cu^{2+} is:

$$\alpha = \frac{c(\text{Cu}^{2+})}{[\text{Cu}^{2+}]} = 1.0 \times 10^8.$$

According to the side reaction coefficient formula:

$$\alpha = 1 + \beta_1[\text{NH}_3] + \beta_2[\text{NH}_3]^2 + \beta_3[\text{NH}_3]^3 + \beta_4[\text{NH}_3]^4 + \beta_5[\text{NH}_3]^5 = 1.0 \times 10^8,$$

the free $[\text{NH}_3]$ is determined to be $[\text{NH}_3] = 4.5 \times 10^{-2} \text{ mol/L}$.

CHECKPOINT

1 PTS

Free $[\text{NH}_3] = 4.5 \times 10^{-2} \text{ mol/L}$

From the distribution coefficient:

$$[\text{NH}_3] = c(\text{NH}_4^+) \cdot \frac{K_a(\text{NH}_4^+)}{[\text{H}^+] + K_a(\text{NH}_4^+)},$$

the concentration $c(\text{NH}_4^+)$ is solved as $c(\text{NH}_4^+) = 9.8 \times 10^{-2} \text{ mol/L}$.

Additionally, the amount of ammonia bound to Cu can be calculated using the complex distribution coefficient:

$$c_{\text{Cu}}(\text{NH}_3) = [\text{Cu}^{2+}] \cdot (\beta_1[\text{NH}_3] + 2\beta_2[\text{NH}_3]^2 + 3\beta_3[\text{NH}_3]^3 + 4\beta_4[\text{NH}_3]^4 + 5\beta_5[\text{NH}_3]^5) = 3.91 \times 10^{-3} \text{ mol/L}.$$

Thus, the total NH_3 concentration is 0.102 mol/L .

CHECKPOINT

2 PTS

The total NH_3 concentration is 0.102 mol/L

From this, the amount of ammonia bound to Cu accounts for approximately 0.038 of the total ammonia, which can be neglected in terms of its consumption during solid dissolution.

CHECKPOINT

1 PTS

The amount of ammonia bound to Cu accounts for approximately 0.038 of the total ammonia

In summary, the analytical concentration of ammonium carbonate is half of the total NH_3 concentration. Thus, the analytical concentration of ammonium carbonate and the minimum mass of ammonium carbonate to be added are calculated as 0.051 mol/L and 0.49 g, respectively.

CHECKPOINT

1 PTS

The analytical concentration of ammonium carbonate is 0.051 mol/L

CHECKPOINT

2 PTS

The minimum mass of ammonium carbonate to be added is 0.49 g